

MEC E 686 Assessment and Analysis of Biomechanical Motion



Winter 2026

Instructor: Dr. Hossein Rouhani

Assignment 1 (100 marks)

Important assignment information: This assignment is due on **Thursday, March 5, 2026**. Please submit a single PDF file of your assignment solution via the Canvas course webpage before 11:00 am on this date. In completing the assignment, **students must provide complete solutions:** the process used in answering the assignment questions should be described, and any MATLAB code developed as part of the solution process should be referenced and included in an appendix.

The following files have been included with this assignment:

MecE686_Assignment1_Guide.pdf: Contains the equations and three-dimensional (3D) joint angle sign conventions required to answer the assignment questions.

MC_P_#_T_#.txt: Time series for the motion capture data recorded during the walking trials. Marker coordinates were recorded in units of mm. Each file contains the data for a single trial. For example, MC_P_2_T_4.txt contains the data for the fourth trial of Participant 2. The 'Frame' column labels the motion capture samples (e.g., since the fourth trial of Participant 2 has 652 samples, the trial length was 6.51 seconds). The 'Sub-Frame' column relates to the force plate and electromyogram recordings and should be ignored in this assignment. The anatomical landmark abbreviations correspond to those defined in Table 1 of the Biomechanics Assignments Overview document, with left and right body side landmarks denoted by 'L' and 'R', respectively, preceding the landmark abbreviation (e.g., RCA for right calcaneus). The force plate markers are abbreviated as FP1, FP2, FP3, and FP4.

Using the information contained in the included files as well as the Assignments Overview document, answer the following questions:

1. **(15 marks)** For each trial and motion capture sample, calculate the rotation matrix (i.e., the 3-by-3 matrix, R) associated with the local coordinate system (LCS) of the pelvis, left and right thigh, left and right shank, and left and right foot (*this should be done according to the equations provided in the Biomechanics Assignment 1 Guide*). **Note that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
2. **(10 marks)** Using the results of Question 1, for each trial, motion capture sample, and body side, calculate the Helical Angles of the hip, knee, and ankle in units of degrees (*this should be*

done according to the equations provided in the Assignment 1 Guide). **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**

3. **(15 marks)** Using the results of Question 1, for each trial, motion capture sample, and body side, use the Joint Coordinate System approach to calculate the 3D angles (i.e., flexion/extension (F/E), adduction/abduction (A/A), and external-rotation/internal-rotation (ER/IR)) of the hip, knee, and ankle in units of degrees (*this should be done according to the equations and sign conventions provided in the Biomechanics Assignment 1 Guide*). **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
4. **(15 marks)** For each trial, determine: (1) the body side of the foot that landed on the force plate; (2) the sample corresponding to this force plate heel (i.e., calcaneus) strike; and (3) the sample corresponding to the next heel strike (i.e., following the force plate heel strike) of the same foot. This information is provided for two trials in Table 1 at the end of this document. **Find the remaining entries of Table 1 and provide its completed version in your solution report. Additionally, the process used to obtain these results should be described, and associated MATLAB code should be referenced and included in an appendix.**

Hints: The heel strike samples for each trial can be determined from the heel marker Z-coordinate trajectories: Heel strikes occur at relative minima for the Z-coordinate (i.e., the coordinate in the vertical direction). During a force plate landing, the heel marker is inside the rectangle formed by the force plate markers; hence, the heel marker Y-coordinate (i.e., the coordinate in the direction of walking) is between the Y-coordinates of the force plate markers during a force plate landing.

In Figure 1 at the end of this document, the Z-coordinate (top panel) and Y-coordinate (bottom panel) trajectories of the left and right heel markers are shown for the first trial of Participant 1. Also shown are the Y-coordinate trajectories of the force plate markers for the same trial (bottom panel). The force plate heel strike sample (sample 332; green diamond) and next heel strike sample (sample 459; black diamond) are indicated on the Z-coordinate trajectory of the right heel marker. Note that, since the sides of the force plate were aligned with the XY-plane of the laboratory coordinate system, only two of four force plate marker Y-coordinate trajectories are distinguishable.

5. **(5 marks)** Using the results of Questions 2, 3, and 4, for each trial, extract the helical and 3D joint angles for the force plate gait cycle (a gait cycle is the heel strike to the following heel strike of the same foot) of the body side of the foot that landed on the force plate. For example, for the first trial of Participant 1, you should extract the angles of the right body side from samples 332 to 459 and discard the rest of the data for this trial (i.e., the remaining right body side data and all the left body side data for this trial should be discarded). **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be**

referenced and included in an appendix.

6. **(5 marks)** Note: For this question and the following ones, analyze only the data extracted in Question 5. Using the results of Question 5, use linear interpolation to time normalize the helical and 3D joint angle gait cycle time series for each trial to 101 samples (i.e., such that the first and last samples of each normalized time series correspond to 0 and 100% of the gait cycle). This should be done using the MATLAB command `interp1(unts,linspace(1,ol,nl))`, where `unts` is an unnormalized time series with length `ol` (in units of samples) and `nl` is the normalized time series length (i.e., 101 samples). **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
7. **(10 marks)** Using the results of Question 6, for each participant, calculate the ensemble average (i.e., mean and standard deviation (SD)) helical and 3D joint angle time series for each body side separately. For example, the ensemble average F/E angle of the left hip of Participant 1 should be calculated from **three** time series. **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
8. **(15 marks)** Using the results of Question 7, for each participant, create a 1-by-3 (one column for each joint) figure that presents the ensemble average helical joint angles, as well as a 3-by-3 (one row for each angle and one column for each joint) figure that presents the ensemble average 3D joint angles. The ensemble average angles should be presented as the mean ± 1 SD bounds. Additionally, for a given angle and joint, the ensemble average angles for each body side should be plotted on the same axes. The helical and 3D joint angle figures for Participant 1 are provided in Figures 2 and 3, respectively, to illustrate how the figures should look. **Provide the created figures (six figures total). Based on a visual inspection of the figures, provide a qualitative assessment of the intra-participant repeatability of the experimental procedure. Additionally, the MATLAB code used to create the figures should be referenced and included in an appendix.**
9. **(10 marks)** Starting with the results of Question 6, filter the helical joint angles for Participant 1 using a fourth-order, zero phase-shift, low-pass Butterworth filter with a cut-off frequency of 5 Hz. This should be done using the MATLAB command `filtfilt(num,den,ufts)`, where `num` and `den` are the filter coefficients and `ufts` is an unfiltered time series. The filter coefficients should be calculated using the MATLAB command `[num,den]=butter(fo,fc/(0.5×fs))`, where f_o is the filter order, f_c is the cut-off frequency, and f_s is the motion capture sampling frequency. Using these filtered angles, repeat the ensemble average calculations of Question 7 and recreate the helical joint angle figure of Question 8 for Participant 1 only. **Based on a visual comparison of the unfiltered and filtered helical joint angles of Participant 1, provide a qualitative assessment on how the filtering process has affected the shape of the mean part of the ensemble average joint angles. Additionally, the MATLAB code used to obtain these results should be referenced and included in an appendix.**

Table 1: Force plate (FP) landing information for each trial.

Participant	Trial	FP landing body side	FP heel strike sample	Next heel strike sample
1	1	Right	332	459
1	2			
1	3			
1	4			
1	5			
1	6			
2	1	Left	395	505
2	2			
2	3			
2	4			
2	5			
2	6			
3	1			
3	2			
3	3			
3	4			
3	5			
3	6			

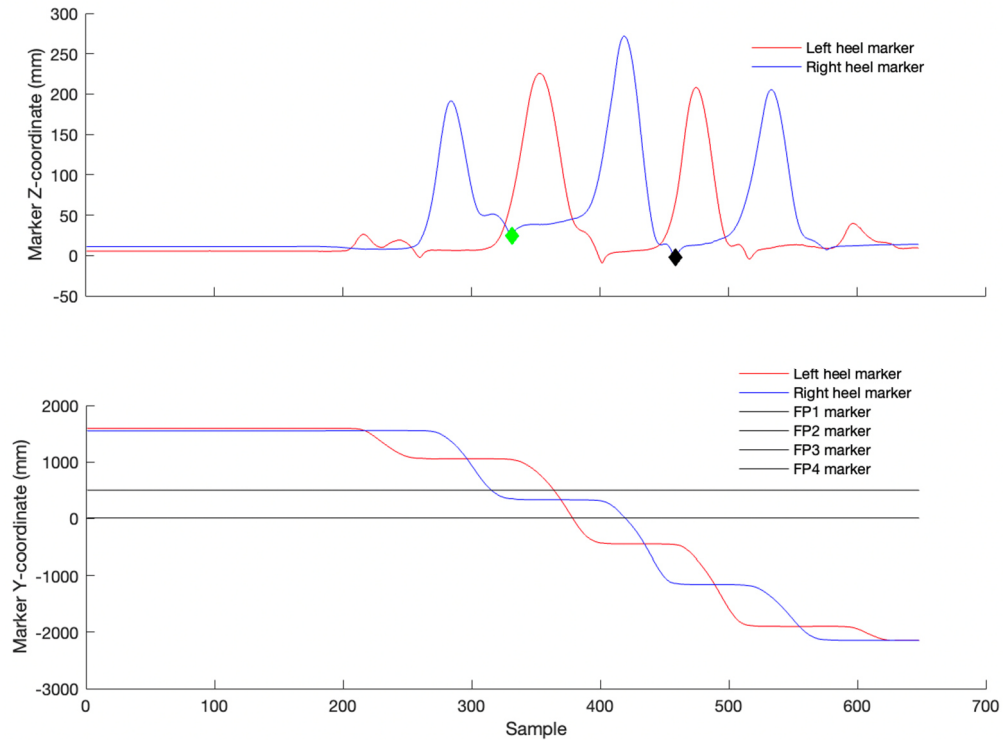


Figure 1: Motion of the heel markers for the first trial of Participant 1. Top panel: Z-coordinate trajectories of the heel markers; bottom panel: Y-coordinate trajectories of the heel and force plate markers. The force plate heel strike sample (sample 332; green diamond) and next heel strike sample (sample 459; black diamond) are indicated on the Z-coordinate trajectory of the right heel marker.

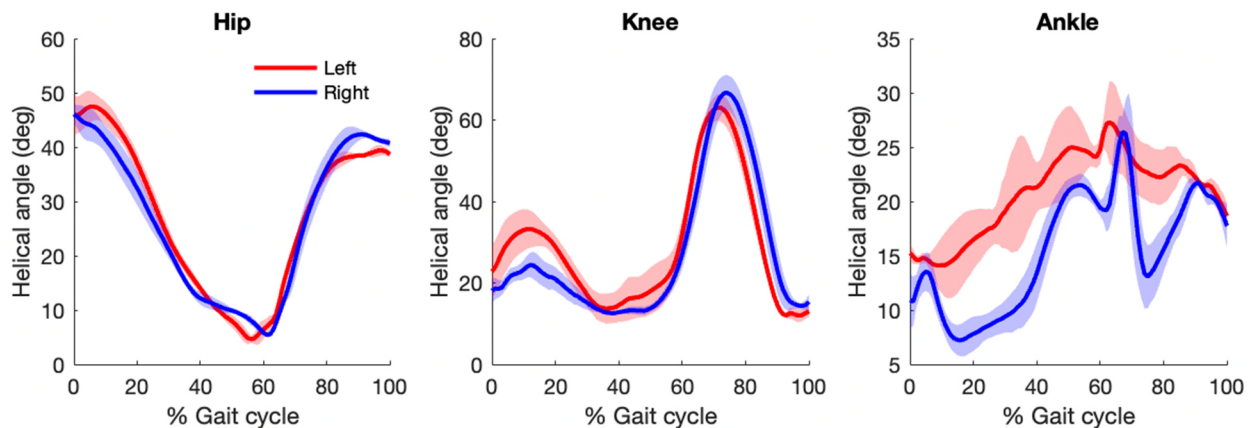


Figure 2: Ensemble average (mean ± 1 standard deviation bound) helical joint angles for Participant 1. Shown are the angles of the left and right hip, knee, and ankle.

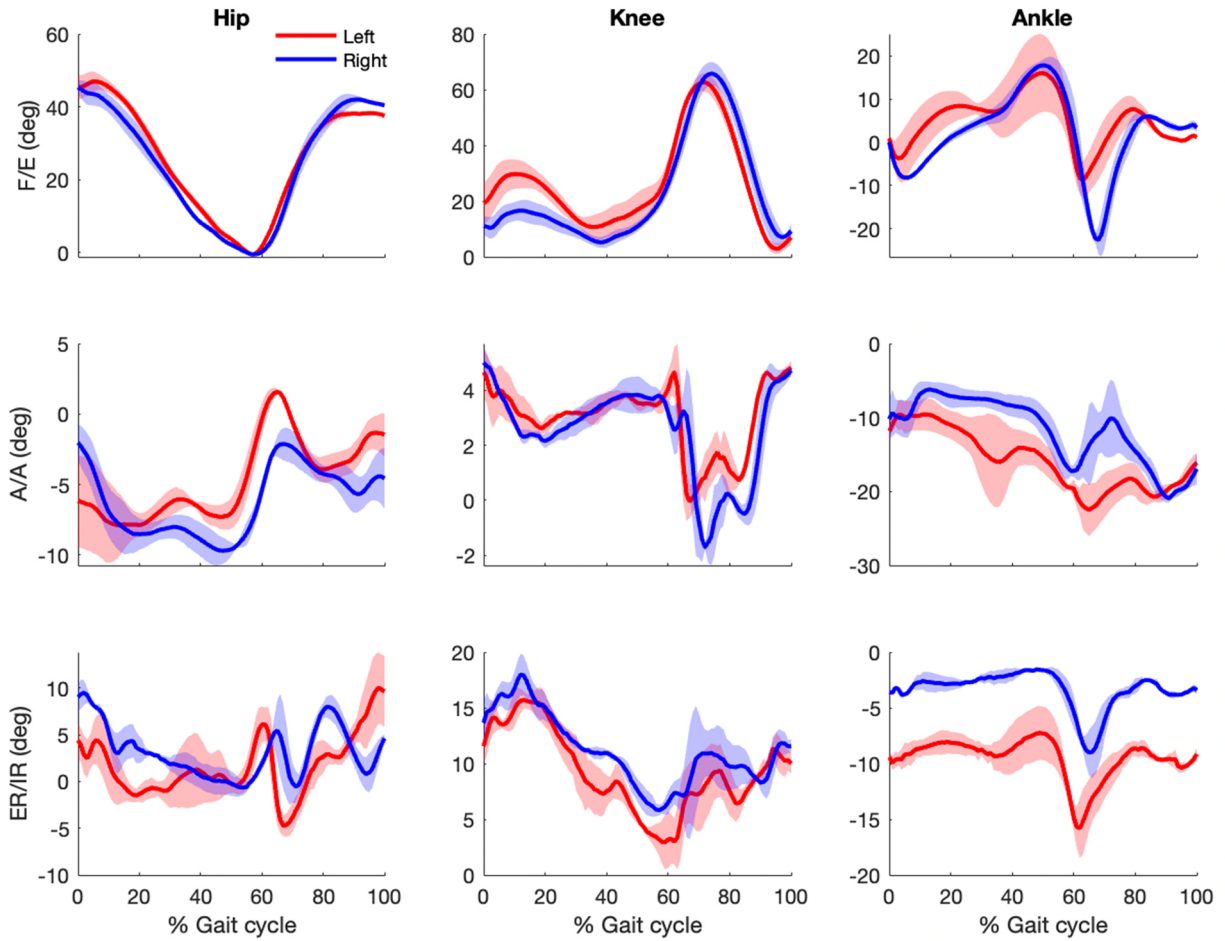


Figure 3: Ensemble average (mean ± 1 standard deviation bound) three-dimensional joint angles for Participant 1. Shown are the flexion/extension (F/E), adduction/abduction (A/A), and external-rotation/internal-rotation (ER/IR) angles of the left and right hip, knee, and ankle.